

KINETIC THEORY OF MATTER

The Kinetic theory of matter assumes that matter is made up of tiny particles such as atoms, molecules and ions that are continually moving and therefore contains kinetic energy. This is a scientific theory that explains the properties and behavior of matter in terms of the motion of its particles. This theory provides insights into how matter exists in different states—solid, liquid, and gas—and how it changes from one state to another.

A British Scientist, Dalton; about 1806AD, expanded the idea of atoms. He made various statements about the nature of atoms. His ideas are:

1. Matter is made up of minute/tiny indivisible particles called atoms.
2. Atoms can neither be created nor destroyed.
3. All atoms of an element are alike and differ from atoms of all other elements.
4. Atoms combine in whole numbers to form new substances.

All these ideas collectively form Dalton's atomic theory. Since Dalton's atomic theory was proposed, Scientists have accepted the idea that matter is made up of particles.

MOLECULAR STRUCTURES OF SOLIDS, LIQUIDS AND GASES.

==>PROPERTIES OF SOLIDS: The following are properties of solids:

- i). They have fixed shapes and cannot flow.
- ii). The particles cannot move from one place to another
- iii). They cannot be compressed or squashed.

==>PROPERTIES OF LIQUIDS:

- i). Liquids flow and usually take the shapes of their containers.
- ii). The particles can move around each other.
- iii). They cannot be compressed or squashed.

==>PROPERTIES OF GASES:

- i). They move and completely fill their containers.
- ii). The particles can move quickly in all directions.
- iii). They can be compressed or squashed.

DIFFERENCES BETWEEN BOILING AND EVAPORATION:

BOILING POINT is the point when liquid compounds start to boil. At this point, the kinetic energy of the particles of the compound increases which leads to the strong vibration between the particles.

While **EVAPORATION** takes place when the liquid particles at surfaces have enough kinetic energy. It is a process where liquids change to gases.

ASSUMPTIONS OF THE KINETIC THEORY

The kinetic theory of matter is based on several key assumptions:

1. **Particles Are in Constant Motion:** The particles are in perpetual motion, with their speed and direction influenced by temperature and other factors.
2. **All Matter is Made of Particles:** Matter consists of a large number of small particles (atoms or molecules) that are constantly moving.
3. **Gaps Between Particles:** There are spaces between the particles, and these spaces can vary significantly between different states of matter. In gases, the spaces are much larger than in liquids and solids.
4. **Negligible Forces at Large Distances:** The forces between particles become negligible at larger distances. In gases, the impact of intermolecular forces can usually be ignored during collisions.
5. **Elastic Collisions:** When particles collide, they do so in perfectly elastic collisions, meaning total energy and momentum are conserved. This assumption simplifies the understanding of gas behavior and pressure.
6. **Average Kinetic Energy:** The average kinetic energy of the particles is directly proportional to the temperature of the substance. When the temperature increases, the average kinetic energy of the particles increases.

ASSIGNMENT

Using Kinetic theory, explain the differences between a solid and a liquid.